The FITWISE

Project Report

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2024329

1. INTRODUCTION

In the contemporary health and wellness landscape, users encounter significant challenges in tracking their physical fitness, nutritional intake, and daily habits seamlessly across various isolated applications. The absence of a unified health assistant contributes to a disjointed user experience, adversely affecting consistency and goal attainment.

In response to this issue, our all-encompassing web application, TheFitWise, serves as an indispensable daily health companion. It seamlessly incorporates features such as an AI-powered meal analyser, comprehensive workout logging, gamified habit tracking, and integrated skincare routines, all consolidated into a single responsive dashboard.

In addition to these capabilities, the latest iteration of TheFitWise significantly evolves beyond passive health tracking systems. While traditional applications focus primarily on logging and visualization, modern user expectations demand real-time intelligence and interactivity.

To address this, TheFitWise now incorporates agentic AI systems for dynamic decision-making, real-time computer vision for exercise analysis, and interactive feedback loops for continuous user improvement. This transformation shifts the application from a static tracking tool to an active AI-powered coaching system, capable of understanding user intent, adapting workouts dynamically, and providing immediate actionable insights.

2. MOTIVATION

Addressing the fragmentation in health tracking is not just a convenience but a critical enhancement to the overall user experience in personal wellness. Due to the lack of a cohesive tool, the issues faced are:

- Fragmented User Experience: Users are forced to switch between separate apps for diet, workouts, and habit tracking, leading to inconsistency.

- Complex Meal Logging: Current dietary applications rely on tedious manual entry and barcode scanning, making it difficult to log complex or home-cooked meals.

- Need for Personalized Assistance: Users lack access to intelligent, context-aware systems that can provide tailored fitness and health recommendations.

- Lack of Real-Time Feedback: Most applications do not provide live guidance during workouts, preventing users from correcting form and improving execution efficiency.

- Static Nature of Existing Systems: Workout plans are often rigid and fail to adapt dynamically based on user performance or goals.

3 TECHNOLOGIES & ALGORITHMS

• Frontend Interface:
The application is built using SvelteKit, enabling a highly performant and modular frontend architecture. Tailwind CSS and shadcn components provide a modern, responsive, and accessible user interface.

• Backend & Database:
Supabase is utilized for PostgreSQL database management, secure user authentication, and serverless backend functionality.

• AI Integration (Agentic System):
The system employs an advanced agentic AI workflow where the model performs multi-step reasoning and interacts with internal tools to dynamically generate personalized workout plans.

• RAG (Retrieval-Augmented Generation):
A vector database is integrated to retrieve relevant fitness knowledge, ensuring responses are grounded, accurate, and context-aware.

• Computer Vision:
MediaPipe is used for real-time pose detection, enabling tracking of body keypoints, joint angle estimation, and movement analysis for exercise validation.

• Algorithms:
The Mifflin-St Jeor equation is implemented to calculate daily caloric requirements based on user-specific parameters.

4 NOVELTY

In comparison to existing systems, TheFitWise offers a significantly enhanced user experience by transitioning from passive tracking to active coaching.
A key feature remains the AI Meal Analysis, which allows users to describe meals in natural language, reducing friction in dietary logging.
Furthermore, the updated system introduces real-time interactivity, enabling dynamic workout generation through agentic AI and immediate feedback via pose detection. The integration of reasoning-based AI and computer vision creates a highly immersive and effective fitness experience.
Additionally, the platform combines multiple modalities—text-based interaction, visual analysis, and structured health data—into a unified system, which is uncommon in current consumer fitness applications.

5 RELATED WORK

Current products in the market, such as MyFitnessPal and Strong, serve as generalized tools for diet and workout tracking. However, they operate in silos and lack real-time feedback and intelligent coaching capabilities.
Previous systems in diet tracking rely heavily on exact-match database queries, which struggle with unstructured inputs. Recent AI-based systems introduce conversational interfaces but lack integration with real-time computer vision and adaptive reasoning.
TheFitWise bridges this gap by integrating agentic AI, real-time feedback, and a unified user experience into a single platform.

6 PIPELINE & DATABASE ARCHITECTURE

The backend architecture of TheFitWise is designed to process both structured and unstructured data efficiently.

6.1 Database Schema

The system uses a PostgreSQL database hosted on Supabase with tables such as:

- profiles
- meals & meal_items
- workouts
- user_stats
- weight_logs, water_logs, caffeine_logs

6.2 AI Meal Analysis Pipeline

1. User inputs meal description
2. Request processed via backend
3. LLM extracts structured nutritional data
4. Output stored and displayed

6.3 Real-Time Feedback Pipeline

1. Video input captured via browser
2. MediaPipe extracts skeletal keypoints
3. Joint angles are computed
4. Movement is evaluated against correct form
5. Feedback generated:
    - Score
    - Corrections
    - Suggestions

6.4 Agentic AI Pipeline

1. User provides fitness goal
2. AI decomposes task into sub-steps
3. Exercises are selected dynamically
4. Plan is generated and refined
5. UI presents actionable workout

6.5 System Interaction Flow

The system operates as a closed-loop architecture:
User Input → AI Processing → Feedback → User Adjustment → Continuous Improvement

7 GAMIFICATION EVALUATION

7.1 Gamification Algorithms

The system awards Experience Points (XP) based on user actions such as logging activities and completing workouts. Levels and streaks are dynamically calculated to maintain engagement.
In the updated system, real-time performance scores from pose detection can also contribute to gamification, rewarding quality of execution.

7.2 Evaluation & Results

• Performance: Real-time pose detection operates efficiently within browser constraints.
• AI Effectiveness: Agentic workflows generate more adaptive and personalized workout plans.
• User Experience: Real-time feedback significantly improves engagement and usability.

8 POTENTIAL CONTRIBUTIONS

This project contributes a scalable architecture for next-generation AI-driven fitness systems.
Key contributions include:

• Integration of agentic AI in consumer applications
• Real-time computer vision-based feedback
• Combination of LLMs, RAG, and CV in one system
• A blueprint for interactive AI-powered coaching systems

TheFitWise demonstrates how intelligent systems can transform health tracking into a proactive, engaging, and personalized experience.